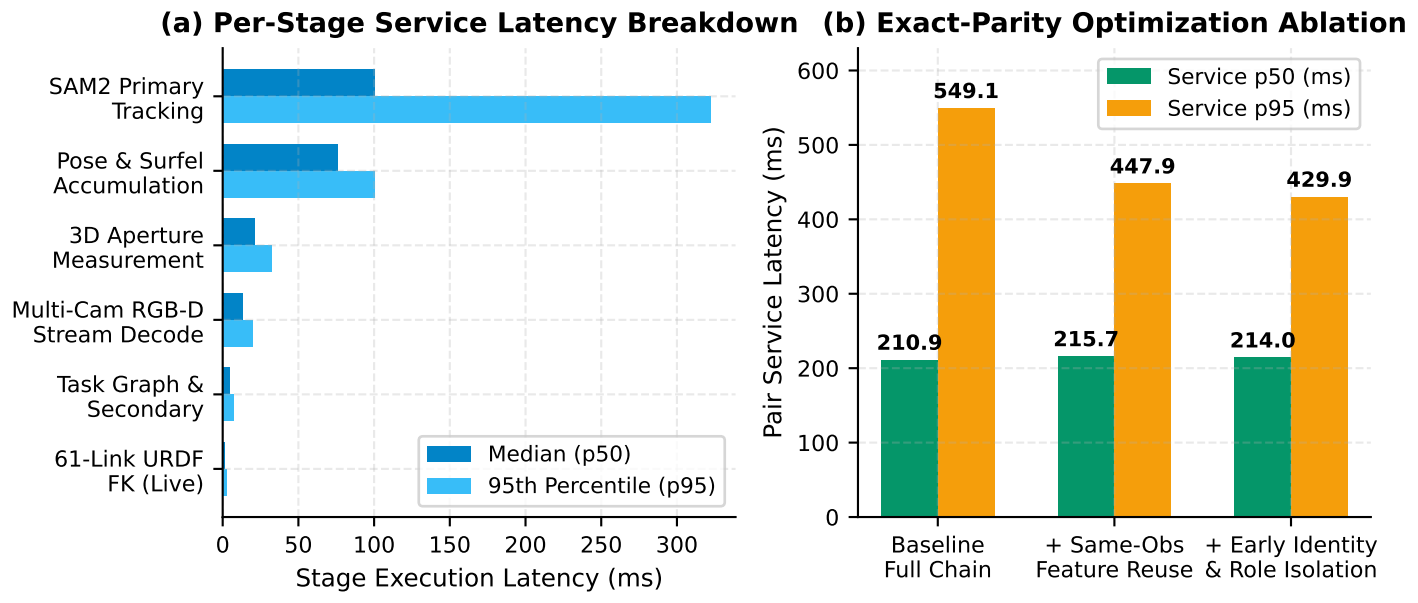
(a) Per-Stage Service Latency Breakdown
SAM2 Primary Tracking
Pose & Surfel Accumulation
3D Aperture Measurement
Multi-Cam RGB-D Stream Decode
Task Graph & Secondary
61-Link URDF FK (Live)
Median (p50)
95th Percentile (p95)
0
50
100
150
200
250
300
Stage Execution Latency (ms)
(b) Exact-Parity Optimization Ablation
Pair Service Latency (ms)
Service p50 (ms)
Service p95 (ms)
600
500
400
300
200
100
0
549.1
447.9
429.9
210.9
215.7
214.0
Baseline Full Chain
+ Same-Obs Feature Reuse
+ Early Identity & Role Isolation